

A PSP Subcase of the Property (K) Unconditional-Basis Question

Literature-implied answer packet for `fa_banach_001`

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Status

This packet records a *literature-implied partial subcase* of an open problem from Aviles–Rodriguez [1]. It is not a full solution of the problem and not an original proof from this run.

Source Problem

Aviles and Rodriguez ask the following as Problem 2.18 of [1]:

Let X be a Banach space with unconditional basis not containing subspaces isomorphic to c_0 . Does X have property (K)?

The original problem appears together with two other final open questions:

THE SPACE $\ell^p(\ell^1, \ell^1)$ HAS THE PROPERTY (K) —

From the previous theorem it follows that $\ell^p(\ell^1)$ has property (K) whenever $1 < p < \infty$. However, this space does not embed isomorphically into any SWCG Banach space, see [16, Corollary 2.29].

We finish the paper with some open questions:

Problem 2.16. *Given a Banach space X and a subspace $Y \subseteq X$, does X have property (K) if both Y and X/Y have property (K)?*

Problem 2.17. *What is the “optimal” cardinal for Theorem 2.14 to work?*

As we already mentioned in the introduction, the conclusion of Theorem 2.14 might fail if $|I| = \mathfrak{b}$.

Problem 2.18. *Let X be a Banach space with unconditional basis not containing subspaces isomorphic to c_0 . Does X have property (K)?*

A natural candidate to test the previous question is the Banach space $E_{1,2}(\mathcal{A})$

Here property (K) means that every weak-star null sequence in X^* admits a convex block subsequence converging to 0 in the Mackey topology $\mu(X^*, X)$.

Supporting Literature

Rodriguez [2] introduced the stronger property (μ^s) . The paper records the implication

$$(\mu^s) \implies \text{property (K)}.$$

$$v_k := \frac{1}{2^k} \sum_{n=1}^{2^k} u_n \quad \text{and} \quad w_k := \frac{1}{2^{k-1}} \sum_{n=2^{k-1}+1}^{2^k} u_n$$

for every $k \in \mathbb{N}$. Then $(w_k)_w$ is a convex block subsequence of $(u_n)_n$ converging to 0, because $(v_k)_k$ converges to 0 and $v_k = \frac{1}{2}(v_{k-1} + w_k)$ for all $k \geq 2$. $\square$

In particular, *property (μ^s) implies property (K)*. The converse is not true in

In the Banach-lattice applications, Rodriguez proves that every Banach lattice with the positive Schur property (PSP) and a weak unit has property (μ^s) :

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A Banach lattice is said to have the *positive Schur property (PSP)* if every weakly null sequence of positive vectors is norm null. Obviously, this property is satisfied by the L^1 -space of any non-negative measure. The PSP implies weak sequential completeness (see e.g. [30, Theorem 2.5.6]) and so the order continuity of the norm. On the other hand, it is known that the PSP is equivalent to saying that every relatively weakly compact set is L -weakly compact (see e.g. [20, Theorem 3.14]). So, Theorem 5.1 implies that *every Banach lattice with the PSP and weak unit has property (μ^s)* . In fact, such a Banach lattice is S^2 WCG, as we show in Corollary 5.4 below. The key is the following lemma (cf. [18, Corollary 5.6]):

Proof Intuition

The 2016 problem is difficult because an unconditional basis alone need not make the Mackey topology on X^* sequentially manageable. The extra PSP hypothesis gives a lattice mechanism: in a Banach lattice with PSP, weakly compact sets are L -weakly compact, and Rodriguez's theorem turns that structure into the stronger Mackey-Cesaro property (μ^s) . A countable unconditional basis supplies a natural coordinate lattice with a weak unit, so the supporting theorem applies directly to this PSP subfamily.

Theorem

Theorem 1. *Let X be a Banach space with an unconditional Schauder basis. Equip X with the associated coordinate Banach-lattice order. Suppose this Banach lattice has the positive Schur property. Then X has property (K).*

Consequently, among spaces satisfying the hypotheses of Aviles–Rodriguez Problem 2.18, the subcase in which the associated unconditional-basis lattice has the positive Schur property has an affirmative answer.

Proof. Let (e_n) be the unconditional basis. After replacing e_n by $e_n/\|e_n\|$, if necessary, the series

$$u = \sum_{n=1}^{\infty} 2^{-n} |e_n|$$

converges in X . In the coordinate lattice associated with the unconditional basis, u is a weak unit: if $x \geq 0$ and $x \wedge u = 0$, then every coordinate of x is zero, hence $x = 0$.

By the additional hypothesis, this Banach lattice has the positive Schur property. Rodriguez's Banach-lattice application [2, Section 5] therefore gives property (μ^s) for X . By [2, Lemma 2.1 and the following sentence], property (μ^s) implies property (K). Thus X has property (K).

If X also contains no subspace isomorphic to c_0 , then X lies in the original class considered in Problem 2.18. The conclusion above is therefore an affirmative answer for that PSP subcase. $\square$

Verification Notes

The supporting paper does not explicitly say that it answers Problem 2.18 of [1]. The identification is direct: an unconditional basis gives a coordinate Banach-lattice structure with a weak unit, and the supporting PSP theorem gives a property stronger than (K).

The PSP assumption is strictly additional. This packet leaves open the full question for arbitrary Banach spaces with unconditional basis and no copy of c_0 .

Bounded Search

The run indexes were searched for the arXiv ids, property (K), positive Schur property, PSP, weak unit, unconditional basis, and c_0 . Web searches using exact phrases from the three final open questions in [1] found later property (K) papers but no full answer to Problem 2.18. The supporting result used here is from arXiv:1812.10079.

References

- [1] A. Aviles and J. Rodriguez, *Convex combinations of weak*-convergent sequences and the Mackey topology*, Mediterr. J. Math. 13 (2016), 4995–5007. arXiv:1601.05825.
- [2] J. Rodriguez, *Cesaro convergent sequences in the Mackey topology*, Mediterr. J. Math. 16 (2019), Paper No. 117. arXiv:1812.10079.